

The Bottle–Cap Marble Run

A bit of a stretch · Indoors · Grown-up nearby · Ages 6–10 · 40 min

Brought to you by Mateo — “If it can be stacked, it can be a city.”

Tape a run down a wall and get the marble to the bottom without it ever leaving the track.

WHAT YOU NEED

- Cardboard tubes and strips
- Bottle caps
- Masking tape
- A marble or a large dried bean
- A wall, door, or big box to build on

BEFORE YOU START

Masking tape only, so the wall survives.

Marbles are a choking hazard. Keep them away from anyone under three, and collect them all at the end.

How to make it

- 1 Tape one ramp near the top of your wall, tilted gently downward.
- 2 Drop the marble. Note exactly where it leaves the ramp and lands.
- 3 Tape the next ramp so it starts where the marble landed, tilted the other way.
- 4 Keep going down the wall, alternating direction. Add walls of folded card wherever the marble flies off.
- 5 Make one ramp steeper than the rest and watch what it does to the speed.
- 6 Put a cup at the bottom and run it five times. Count how many times it lands in the cup.

TALK ABOUT IT

Where did your marble keep escaping, and what finally kept it on the track?

TRY THIS TOO

Add a jump: a gap the marble has to fly across into a funnel. Adjust the ramp before it until it works.

DID YOU KNOW? · SHAPES & STRUCTURES

A steeper ramp gives a faster roll. Gravity pulls straight down all the time. A steep ramp lets more of that pull go into speeding the marble along, so it picks up speed faster. A gentle ramp turns the same drop into a longer, slower journey.